Research Discussion Notes: Picard Groups, Lattices, and Partitions

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1 Picard Group via the Incidence Matrix

Question 1. *What happens if we define the Picard group using the incidence matrix rather than the Laplacian?*

Discussion 1. *The incidence matrix B is $n \times m$ (vertices $\times$ edges). The Laplacian is $\Delta = BB^T$.*

If we define the "edge Picard group" as $\mathbb{Z}^E/\operatorname{im}(B^TB)$: B^TB is the $m \times m$ edge Laplacian. Its kernel is the cycle space (dimension $g = m-n+1$). Its image is in the cut space (dimension $n-1$).

The nonzero eigenvalues of B^TB and BB^T are identical. But $\mathbb{Z}^E$ has dimension m, while $\operatorname{im}(B^TB)$ has rank $n-1$. So the quotient $\mathbb{Z}^E/\operatorname{im}(B^TB)$ is infinite — it has a free part of rank $m-(n-1) = g+1$ plus a torsion part isomorphic to $\operatorname{Pic}^0(G)$.

If we restrict to the cut space, we recover $\operatorname{Pic}^0(G)$ exactly. Other constructions using B directly give either trivial groups ($\mathbb{Z}^V/\operatorname{im}(B) \cong \mathbb{Z}$) or infinite groups ($\mathbb{Z}^E/\operatorname{im}(B^T)$).

The Laplacian is the only construction that yields a finite nontrivial group from a graph.

2 The Quotient $\mathbb{Z}^E/(\text{Cut Lattice} + \text{Cycle Lattice})$

Question 2. *What about $\mathbb{Z}^E/(\operatorname{im}(B^T) + \ker_{\mathbb{Z}}(B))$?*

Discussion 2. *Over $\mathbb{R}$: cut space $\oplus$ cycle space $= \mathbb{R}^E$. Every edge vector decomposes uniquely.*

Over $\mathbb{Z}$: $\operatorname{im}(B^T) + \ker_{\mathbb{Z}}(B)$ might not equal $\mathbb{Z}^E$. However, for graphs with the standard incidence matrix, the fundamental cycles and cocycles (with respect to any spanning tree)

form a unimodular *basis of* $\mathbb{Z}^E$*. This is because the graphic matroid is regular, and regular matroids are representable by totally unimodular matrices. Therefore:*

$$\mathrm{im}(B^T) + \ker_{\mathbb{Z}}(B) = \mathbb{Z}^E.$$

The quotient is trivial. No new group.

This unimodularity is special to graphic matroids. For non-regular matroids, the quotient could be nontrivial — this is related to the work of Baker–Ding–Zhuang on sixth-root-of-unity matroids.

3 Chip-Firing as Flow

Question 3. *Chip-firing at vertex v feels like a flow of all chips at v to its neighbours equally. Is this correct?*

Discussion 3. *Yes, that is exactly what it is. Firing vertex v:*

- *v loses* $\deg(v)$ *chips (one per edge),*
- *each neighbour of v gains 1 chip.*

It is a unit flow pushed out equally along every edge incident to v. The total flow out of v equals $\deg(v)$*. Each edge carries exactly 1 unit.*

This is why $\Delta_G = BB^T$: the incidence matrix B routes flow along edges, and B^T aggregates it at vertices. Chip-firing at v is B applied to the edge flow pattern "1 unit on every edge of v," which is $B^T(v, \cdot)$.

The key limitation: you can only fire all edges of v simultaneously with equal flow. You cannot choose to send 2 chips along one edge and 0 along another. This rigidity is what makes the quotient $\mathbb{Z}^V / \mathrm{im}(\Delta)$ finite — the chip-firing moves are "too coarse" to reach every integer configuration.

4 Flow Methods and the Picard Group

Question 4. *Can we implement flow methods with the Picard group?*

Discussion 4. *The connection is direct:*

- *Chip-firing at v = push 1 unit of flow out of v along every edge.*
- *Chip-firing at a set S of vertices = push flow out of every vertex in S simultaneously.*
- *The characteristic script $\chi(V_i)$ from Xu's thesis = fire all vertices in district V_i = push flow out of an entire district across its boundary.*

From Xu's Corollary 4: firing $\chi(V_1)$ on the partition divisor D^S rotates the ordered partition. The flow interpretation: pushing all flow out of V_1 simultaneously redistributes it to neighbouring districts, which cyclically relabels the partition.

The chip-firing / flow duality gives us:

- ***Single vertex fire** = local flow (one node pushes to neighbours).*

- ***District fire*** $\chi(V_i)$ *= global flow (entire district pushes across boundary).*
- ***Difference*** $D^S - D^{S'}$ *between two partition divisors = net flow needed to transform one partition into another (Xu's Corollary 3: this equals the boundary edges between merged districts).*

However, chip-firing moves chips (divisor values), not population. Population is fixed on nodes. The divisor D^S *records boundary degrees, not populations. To connect to population, one needs a divisor encoding population information, such as* $D_{\text{pop}} = p - \bar{p} \cdot \mathbf{1}$.

5 Changing the Divisor Definition

Question 5. *If we change the divisor definition according to population flows, how could the Picard group change?*

Discussion 5. *The Picard group does not change.* $\text{Pic}^0(G) = \mathbb{Z}^V / \text{im}(\Delta_G)$ *depends only on the graph* G*, not on what the divisor values "mean." The group structure is fixed by the Laplacian.*

What changes is which divisor you are looking at within the fixed group. D^S *and* D_{pop} *are two different elements of* $\text{Div}(G)$*. Chip-firing acts on both via the same group action.*

The key insight: if you start at $D_{\text{pop}} = p - \bar{p} \cdot \mathbf{1}$ *and chip-fire, you trace out the coset* $[D_{\text{pop}}]$ *in* $\text{Pic}^0(G)$*. Every element of this coset is a divisor equivalent to* D_{pop} *— a different distribution of population deviations reachable by chip-firing. The coset has exactly* $t(G)$ *elements.*

The question becomes: *within the coset* $[D_{\text{pop}}]$*, which elements correspond to balanced partitions?*

A balanced partition has $|p(\xi_i) - \bar{p}_{\text{district}}|$ *small for each district. In divisor terms: the divisor values, when summed over each district, should be close to zero. Different elements of* $[D_{\text{pop}}]$ *redistribute the population deviations differently — some redistributions might create balanced partitions, others will not.*

So the problem becomes: navigate within a single coset of $\text{Pic}^0(G)$ *to find divisors whose district sums are balanced. The MCMC walks within this coset via chip-firing, and the partition emerges from the divisor values.*

6 Why Chip-Firing Finds Spanning Trees

Question 6. *Why and how is firing a chip useful in finding spanning trees?*

Discussion 6. *Start from the sandpile picture. You have a configuration of chips on vertices. Fix a sink* q*. Keep toppling unstable vertices (those with chips* $\geq$ *degree) until everything is stable. The final stable configuration is unique — this is the* ***abelian property*** *of the sandpile (Dhar, 1990).*

Each recurrent stable configuration corresponds to exactly one spanning tree. There are $t(G)$ *of each. The bijection is concrete: when you run Dhar's burning algorithm from the sink* q*, the edges that "carry the fire" form the spanning tree.*

You are at stable configuration c *(corresponding to tree* T*). You add one chip to some vertex* v*. This might make* v *unstable. You topple* v *— sending one chip to each neighbour.*

This might make a neighbour unstable. You cascade topplings until stable again. You arrive at a new stable configuration c' (corresponding to tree T').

The map $c \to c'$ is deterministic. By choosing different vertices v to add chips to, you systematically explore the space of spanning trees.

The exploration has algebraic structure: adding a chip at v and stabilising corresponds to adding the group element $[\Delta_G(v, \cdot)]$ in $\mathrm{Pic}^0(G)$. The group is abelian, so the order does not matter. After firing vertices $v_1, v_2, \ldots, v_r$, you have added $[\Delta_G(v_1, \cdot)] + \cdots + [\Delta_G(v_r, \cdot)]$ in the group. The resulting tree depends only on this sum, not on the order.

7 Configurations and the Picard Group

Question 7. *Can I say each configuration is an element in the Picard group? Also, you said the final stable configuration is unique. What is the name of this theorem?*

Discussion 7. *Almost. Each* recurrent *stable configuration corresponds to a unique element of* $\mathrm{Pic}^0(G)$*. Not every configuration — only the recurrent ones. There are $t(G)$ recurrent configurations, $t(G)$ elements of* $\mathrm{Pic}^0(G)$*, and $t(G)$ spanning trees. These three sets are in bijection.*

Non-recurrent (transient) configurations do not correspond to group elements. They appear only during the transient phase and never return.

The confluence of topplings is called the ***abelian property*** *of the sandpile model, proved by Dhar (1990). The key statement: if multiple vertices are unstable, the order in which you topple them does not affect the final result. This is why the model is called the* abelian *sandpile model.*

The abelian property is also why the recurrent configurations form a group — the group operation (add two configurations, then stabilise) is well-defined because stabilisation is unique regardless of toppling order.

Reference: D. Dhar, "Self-organized critical state of sandpile automaton models," Physical Review Letters*, 1990.*

8 Natural Connection Between Sandpile Group and Spanning Trees

Question 8. *Is there a natural connection between the sandpile group and the set of spanning trees other than cardinality? Is it just finding a bijection map, or is there a deeper connection?*

Discussion 8. *Deeper. The bijection is not an accident of counting — it comes from the structure of the Laplacian.*

The Laplacian connects three objects simultaneously:

1. ***Spanning trees:*** $\det(\Delta^{(q)}) = t(G)$ *counts them (Kirchhoff).*
2. ***Sandpile group:*** $\mathbb{Z}^{n-1}/\operatorname{im}(\Delta^{(q)})$ *defines it.*
3. ***Chip-firing dynamics:*** $\Delta_G(v, \cdot)$ *is the toppling rule.*

These are not three separate facts. They are three faces of one object: the integer lattice $\mathrm{im}(\Delta^{(q)})$ *inside* $\mathbb{Z}^{n-1}$.

The reduced Laplacian $\Delta^{(q)}$ *is an integer matrix. Its columns generate a sublattice* L *of* $\mathbb{Z}^{n-1}$. *The fundamental domain of* L *(a parallelotope) has volume* $\det(\Delta^{(q)}) = t(G)$. *The lattice points of* $\mathbb{Z}^{n-1}$ *inside one fundamental domain are the representatives of* $\mathbb{Z}^{n-1}/L = \mathrm{Pic}^0(G)$. *There are exactly* $t(G)$ *of them.*

Each lattice point in the fundamental domain is *a chip configuration. The spanning tree bijection shows that this canonical representative encodes a spanning tree.*

The bijection exists because spanning trees are bases of the same lattice that defines the Picard group. *The lattice is* $\mathrm{im}(\Delta^{(q)})$. *Its bases are spanning trees. Its cosets are group elements. The bijection maps each coset to the basis that defines its canonical representative.*

9 Partitions in the Lattice

Question 9. *Then we may need to understand what partitions are in the lattice.*

Discussion 9. *A spanning tree* T *is a basis of the lattice* $\mathrm{im}(\Delta^{(q)})$. *It has* $n-1$ *edges, giving* $n-1$ *independent lattice vectors.*

A partition ξ *removes* $k-1$ *edges from* T, *leaving a forest with* $n-k$ *edges. That forest is a* ***partial basis*** *—* $n-k$ *of the* $n-1$ *lattice vectors.*

So a partition is a ***rank-***$(n-k)$ ***sublattice*** *of* $\mathrm{im}(\Delta^{(q)})$. *Different partitions correspond to different sublattices of the same rank. Two spanning trees that give the same partition share the same* $n-k$ *sublattice vectors (the within-district edges) and differ only in the remaining* $k-1$ *vectors (the boundary edges).*

In lattice terms:

- $\mathrm{Pic}^0(G) = \mathbb{Z}^{n-1}/L$ *where* $L = \mathrm{im}(\Delta^{(q)})$, *a rank-*$(n-1)$ *lattice.*
- *A spanning tree = a basis of* L.
- *A partition = a rank-*$(n-k)$ *sublattice of* L.
- *The fibre = all bases extending this sublattice.*
- *Within-district moves = change the basis without changing the sublattice.*
- *Cross-boundary moves = change the sublattice itself.*

The partition space is analogous to the Grassmannian of rank-$(n-k)$ *sublattices of* L.

10 The Lattice Explicitly

Question 10. *Can you help me understand this lattice? Define it explicitly.*

Discussion 10. *Fix a graph* G *with vertices* $\{v_1, \ldots, v_n\}$ *and a sink* $q = v_n$. *The reduced Laplacian* $\Delta^{(q)}$ *is the* $(n-1) \times (n-1)$ *matrix obtained by deleting the last row and column of* Δ_G.

The lattice L is:

$$L = \{\Delta^{(q)} z : z \in \mathbb{Z}^{n-1}\} \subset \mathbb{Z}^{n-1}.$$

The columns of $\Delta^{(q)}$ are the generators. Column i is the chip-fire vector at vertex v_i (restricted to non-sink vertices).

Example: the triangle C_3.

Vertices $\{1, 2, 3\}$, sink $q = 3$. Each vertex has degree 2.

$$\Delta^{(3)} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

Column 1 $= (2, -1)$: fire vertex 1. Column 2 $= (-1, 2)$: fire vertex 2.

The lattice L is all integer combinations of $(2, -1)$ and $(-1, 2)$:

$$L = \{a(2, -1) + b(-1, 2) : a, b \in \mathbb{Z}\}.$$

This is a sublattice of $\mathbb{Z}^2$. Its fundamental domain is a parallelogram with area $= \det(\Delta^{(3)}) = 4 - 1 = 3 = t(C_3)$.

The three cosets of L in $\mathbb{Z}^2$ are $(0, 0) + L$, $(1, 0) + L$, and $(0, 1) + L$. These are the three elements of $\mathrm{Pic}^0(C_3) \cong \mathbb{Z}/3$, each corresponding to one spanning tree.

11 Kirchhoff's Proof and the Laplacian–Tree Connection

Question 11. *How is the reduced Laplacian related to spanning trees? How does Kirchhoff prove it? Are the submatrices with $n - k$ rows related to partitions?*

Discussion 11. ***Kirchhoff's proof:*** *The Laplacian factors as $\Delta = BB^T$ where B is the oriented incidence matrix ($n \times m$). The reduced version: $\Delta^{(q)} = B^{(q)}(B^{(q)})^T$ where $B^{(q)}$ is $(n-1) \times m$.*

By Cauchy–Binet:

$$\det(\Delta^{(q)}) = \sum_{S \subset E,\, |S| = n-1} \det(B_S^{(q)})^2$$

where $B_S^{(q)}$ is the $(n-1) \times (n-1)$ submatrix using columns indexed by edge set S.

The key fact: $\det(B_S^{(q)}) = \pm 1$ if S is a spanning tree, and 0 otherwise. This is because: if S contains a cycle, the columns are linearly dependent; if S is a spanning tree, the columns are independent, and the determinant is ± 1 because B is totally unimodular. Therefore:

$$\det(\Delta^{(q)}) = \sum_{T \text{ spanning tree}} (\pm 1)^2 = t(G).$$

Submatrices and partitions: *Yes. The* ***Matrix-Forest Theorem*** *(Chebotarev and Shamis, 1997) states: for any subset $S \subset V$ with $|S| = r$, the $r \times r$ minor of Δ_G indexed by S equals the number of spanning forests of G where each component tree contains exactly one vertex from $V \setminus S$ as its root. So:*

- *$(n-1) \times (n-1)$ minor $\to$ spanning trees (1 component, 1 root).*
- *$(n-2) \times (n-2)$ minor $\to$ spanning forests with 2 components.*

- $(n-k) \times (n-k)$ *minor* $\to$ *spanning forests with* k *components* $= k$*-partitions (with designated roots).*

This means the lattice structure at rank $n-k$ *directly encodes partitions, just as the full rank* $n-1$ *lattice encodes spanning trees. The determinant of the* $(n-k) \times (n-k)$ *submatrix counts the partitions (with roots), just as* $\det(\Delta^{(q)})$ *counts spanning trees.*

Key question: *The Picard group lives at rank* $n-1$*. Is there a "partition Picard group" at rank* $n-k$*?*

12 The Coset Formulation

Discussion 12 (Summary of the coset formulation)**.** *Combining the insights from the previous sections, we arrive at the following framework:*

State: *a divisor* $D \in [D_{\text{pop}}]$*, the coset of* $D_{\text{pop}} = p - \bar{p} \cdot \mathbf{1}$ *in* $\text{Pic}^0(G)$*.*

Move: *chip-fire at vertex* v*.* $D \to D - \Delta_G(v, \cdot)$*. Stays in the same coset.*

Partition extraction: *from* D*, define districts by grouping vertices according to the population deviation pattern encoded in* D*.*

Reversibility: *chip-firing with symmetric proposal (fire* v *or anti-fire* v *with equal probability) is reversible on the coset. Uniform stationary distribution on the coset, for free.*

Irreducibility: *the walk visits every element of the coset (chip-firing generates* $\text{Pic}^0(G)$*, coset is a single orbit). If the partition extraction rule is surjective, every partition is reachable.*

Population balance is built in. D_{pop} *encodes population. Every element of* $[D_{\text{pop}}]$ *is a redistribution of the same total population deviation. The partition extraction looks for clustered deviations — districts where the sum is near zero.*

Open problems:

1. *Define a partition extraction rule that is well-defined and surjective.*
2. *Prove that the induced chain on partitions is irreducible.*
3. *Show polynomial mixing time.*

13 Submatrices of the Laplacian and Partitions

Question 12. *Does it matter which* $(n-k) \times (n-k)$ *submatrix we take to generate a partition? How is the reduced Laplacian related to spanning trees, and are smaller submatrices related to partitions?*

Discussion 13. *Yes. Different* $(n-k) \times (n-k)$ *submatrices correspond to different choices of which* k *vertices to delete — and the deleted vertices become the roots of the* k *components.*

The ***Matrix-Forest Theorem*** *(Chebotarev and Shamis, 1997) states: for any subset* $S \subset V$ *with* $|S| = k$*, the minor of* Δ_G *obtained by deleting the rows and columns indexed by* S *has determinant equal to the number of spanning forests of* G *with* k *components, where each component is rooted at one of the vertices in* S*. So:*

- $(n-1) \times (n-1)$ *minor* $\to$ *spanning trees (1 root).*

- $(n-2) \times (n-2)$ *minor* $\to$ *2-component rooted forests.*
- $(n-k) \times (n-k)$ *minor* $\to$ *k-component rooted forests = rooted k-partitions.*

Within one minor, forests are grouped by partition. Each partition ξ appears $\prod_i t(G[\xi_i])$ times (different within-district trees, same components). Two rooted forests from the same minor give the same partition iff they have the same connected component vertex sets.

The choice of which k vertices to delete determines which vertices are roots. Different root choices give different counts and different forest sets. This is an asymmetry with the tree case, where deleting any single vertex gives the same count $t(G)$.

14 The Partition Picard Group $\mathrm{Pic}_k(G)$

Question 13. *What would happen if we consider $\mathbb{Z}^{n-k}$/rank-$(n-k)$ sublattice?*

Discussion 14. *This defines the **partition Picard group** — the analogue of $\mathrm{Pic}^0(G)$ at rank $n-k$ instead of rank $n-1$.*

Fix roots $\{q_1, \ldots, q_k\}$. Delete those k rows and columns from Δ_G to get $\Delta^{(\{q_1,\ldots,q_k\})}$, an $(n-k) \times (n-k)$ matrix. Define:

$$\mathrm{Pic}_k(G) = \mathbb{Z}^{n-k} / \operatorname{im}(\Delta^{(\{q_1,\ldots,q_k\})}).$$

Size $= \det(\Delta^{(\{q_1,\ldots,q_k\})})$ = number of rooted spanning forests with k components rooted at $q_1, \ldots, q_k$.

*Each element of $\mathrm{Pic}_k(G)$ corresponds to one rooted spanning forest. Chip-firing makes sense: the columns of $\Delta^{(\{q_1,\ldots,q_k\})}$ are chip-fire vectors at non-root vertices, where chips arriving at any root vertex disappear (they are deleted from the lattice). This is the **multi-sink sandpile**: each district has a designated sink, and chips drain at roots.*

15 Multi-Sink Bernardi Bijection

Question 14. *Could we define a bijection similar to Bernardi's bijection for $\mathrm{Pic}_k(G)$?*

Discussion 15. *The ingredients exist but the construction appears to be new.*

For Bernardi's bijection at rank $n-1$: fix root q, fix ribbon structure. Tour the graph, record break points, get a divisor. The divisor class bijects with a spanning tree.

At rank $n-k$ with k roots $\{q_1, \ldots, q_k\}$: start k simultaneous tours, one from each root q_i, each exploring its own district. Each tour follows the ribbon structure within its territory. When tours meet at boundary edges, the break points record the boundary structure. The tours' territories are *the partition.*

*Dhar's burning also generalises: burn from k sinks simultaneously. Vertices catch fire from whichever sink reaches them first. The fire fronts meet at district boundaries. The burning process defines both the forest (which edges carry fire) and the partition (which sink claimed each vertex). This is a **Voronoi diagram** on the graph with respect to the k root vertices.*

The bijection would be: recurrent configurations of the k-sink sandpile $\leftrightarrow$ rooted spanning forests $\leftrightarrow$ elements of $\mathrm{Pic}_k(G)$.

As far as we are aware, a published multi-sink Bernardi bijection does not exist. This could be a new contribution.

16 Relation Between $\mathrm{Pic}^0(G)$ and $\mathrm{Pic}_k(G)$

Question 15. *What is the relation between the two Picard groups?*

Discussion 16. *There is a natural surjection* $\pi : \mathrm{Pic}^0(G) \to \mathrm{Pic}_k(G)$.

Given a divisor $D \in \mathbb{Z}^{n-1}$ *(chips on all vertices except* q*), project it to* $\mathbb{Z}^{n-k}$ *by forgetting the coordinates at* $q_2, \ldots, q_k$*. This projection respects chip-firing: if* $D \sim D'$ *in* $\mathrm{Pic}^0(G)$*, then* $\pi(D) \sim \pi(D')$ *in* $\mathrm{Pic}_k(G)$*. Therefore:*

$$\mathrm{Pic}_k(G) \cong \mathrm{Pic}^0(G)/\ker(\pi).$$

The kernel $\ker(\pi)$ *consists of divisor classes in* $\mathrm{Pic}^0(G)$ *that become trivial in* $\mathrm{Pic}_k(G)$ *— classes that can be resolved by chip-firing when* $k-1$ *extra sinks are available to absorb chips.*

Combinatorially: $\ker(\pi)$ *contains the within-district spanning tree changes. Two spanning trees in the same* $\ker(\pi)$*-coset map to the same rooted* k*-forest under projection.*

The sizes satisfy:

$$|\mathrm{Pic}^0(G)| = |\ker(\pi)| \cdot |\mathrm{Pic}_k(G)|,$$

i.e., $t(G) = |\ker(\pi)| \cdot$ (*number of rooted* k*-forests*).

17 Properties of $\mathrm{Pic}_k(G)$

Question 16. *Does every such quotient make a group? How do we know it is abelian?*

Discussion 17. *Every quotient* $\mathbb{Z}^{n-k}/\operatorname{im}(\Delta^{(\{q_1,\ldots,q_k\})})$ *is automatically a finitely generated abelian group. It is abelian because it is a quotient of* $\mathbb{Z}^{n-k}$*, which is abelian — a quotient of an abelian group is always abelian.*

The group is finite (hence useful) when $\Delta^{(\{q_1,\ldots,q_k\})}$ *has full rank* $n-k$*. For a connected graph with* $n > k$*, this is guaranteed.*

By the Fundamental Theorem of Finitely Generated Abelian Groups, the Smith normal form of $\Delta^{(\{q_1,\ldots,q_k\})}$ *gives the decomposition:*

$$\mathrm{Pic}_k(G) \cong \mathbb{Z}/d_1 \times \mathbb{Z}/d_2 \times \cdots \times \mathbb{Z}/d_{n-k},$$

where $d_1 \mid d_2 \mid \cdots \mid d_{n-k}$ *are the invariant factors.*

18 Cyclic Factors and Spanning Trees

Question 17. *What does a cyclic group in the decomposition correspond to in the set of spanning trees?*

Discussion 18. *Each cyclic factor* $\mathbb{Z}/d_i$ *corresponds to an independent "direction" through spanning tree space via chip-firing.*

The group $\mathbb{Z}/d_1 \times \mathbb{Z}/d_2$ *is like a* $d_1 \times d_2$ *grid of positions. Each position is a spanning tree. Moving one step in the first direction (adding 1 to the first coordinate) changes the tree by a specific chip-fire combination. Moving* d_1 *steps returns to the same tree. Moving in the second direction is an independent chip-fire that cycles through* d_2 *trees before returning.*

For the triangle C_3: $\mathrm{Pic}^0(C_3) \cong \mathbb{Z}/3$. One cyclic factor, order 3. Three spanning trees arranged in a cycle. Chip-firing at vertex 1 moves one step: $T_1 \to T_2 \to T_3 \to T_1$.

For $\mathrm{Pic}_k(G)$: the same interpretation, but in rooted forest space. Small factors = short loops through few forests. Large factors = long explorations through many forests.

*The **mixing time** of a random walk on $\mathbb{Z}/d_1 \times \cdots \times \mathbb{Z}/d_r$ is dominated by the largest factor: $t_{\mathrm{mix}} = O(d_{\max}^2 \log r)$. So the largest invariant factor of $\mathrm{Pic}_k(G)$ controls how fast the chain explores all partitions.*

19 Generators and Navigation

Question 18. *Then finding a tree corresponding to each cyclic group would be enough to generate the full group? Is it guaranteed that $n-1$ generators visit all spanning trees?*

Discussion 19. *Yes. $\mathrm{Pic}^0(G)$ has rank $n-1$ (exactly $n-1$ invariant factors). So $n-1$ generators suffice to reach every spanning tree from any starting tree. Every element of $\mathrm{Pic}^0(G)$ is a unique integer combination of these $n-1$ generators. For $\mathrm{Pic}_k(G)$: $n-k$ generators suffice to reach every rooted k-forest.*

*A generator of $\mathbb{Z}/d_i$ is not a single chip-fire at one vertex. It is a specific **integer linear combination** of chip-fires at multiple vertices — the Smith normal form mixes the original chip-fire vectors to align with the cyclic factors. The Smith normal form algorithm computes these generators: when $U\Delta^{(q)}V = \mathrm{diag}(d_1, \ldots, d_{n-1})$, the columns of V^{-1} are the aligned generators.*

This gives an algorithm:

1. *Compute $\Delta^{(\{q_1,\ldots,q_k\})}$.*
2. *Smith normal form $\to$ invariant factors $d_1, \ldots, d_{n-k}$ and transformation matrices.*
3. *Extract generators $g_1, \ldots, g_{n-k}$.*
4. *Starting from any rooted forest, reach any other by: compute the target coordinates, apply the corresponding combination of generators.*

No MCMC needed for exploration — you can jump directly to any forest by applying the right group element. MCMC is only needed for sampling from a non-uniform distribution (e.g., favouring population balance).

***The remarkable fact:** the number of generators $(n-k)$ is polynomial in n, but the number of objects reached (rooted k-forests) is exponential. The group structure compresses exponential exploration into polynomial-dimensional navigation.*

20 Risks and Open Questions

The $\mathrm{Pic}_k(G)$ framework is promising but carries several risks:

1. **Root waste:** $\mathrm{Pic}_k(G)$ indexes rooted forests, not unrooted partitions. Two rooted forests with the same districts but different roots are distinct elements. A chip-fire could change roots without changing the partition.

2. **Multi-sink Bernardi bijection:** The explicit bijection between $\mathrm{Pic}_k(G)$ and rooted k-forests via a multi-source touring procedure has not been published. It likely works for planar graphs (where the ribbon structure controls the tours) but needs proof.

3. **Root dependence:** $\mathrm{Pic}_k(G)$ depends on the choice of k root vertices. Different choices give different groups (different sizes, different invariant factors). There is no canonical root choice for redistricting.

4. **Kernel characterisation:** The claim that $\ker(\pi)$ corresponds exactly to within-district moves needs careful verification. Some cross-boundary chip-fires might cancel at the extra sinks.

These are concrete, testable questions that can be resolved by computation on small examples (triangle, 6×6 grid) before attempting general proofs.

21 The Structure of k-Partitions Inside k-Forests

Question 19. *Can we identify the structure of k-partitions inside the space of k-forests, and embed both inside $\mathrm{Pic}^0(G)$?*

Discussion 20. *The group $\mathrm{Pic}_k(G)$ acts on rooted k-forests. The rooted k-forests partition into equivalence classes, where each class consists of all forests giving the same (unrooted) k-partition.*

The fibre of a partition. *For a fixed connected k-partition $\xi = (\xi_1, \ldots, \xi_k)$, the number of rooted k-forests producing ξ is:*

$$|\mathit{fibre}(\xi)| = \prod_{i=1}^{k} |V(\xi_i)| \cdot t(G[\xi_i]),$$

where $|V(\xi_i)|$ accounts for the choice of root within district ξ_i, and $t(G[\xi_i])$ accounts for the choice of spanning tree within the district.

The stabiliser. *Within $\mathrm{Pic}_k(G)$, the elements that map a forest compatible with ξ to another forest compatible with the same ξ form the* stabiliser $\mathrm{Stab}(\xi) \subset \mathrm{Pic}_k(G)$*, with*

$$|\mathrm{Stab}(\xi)| = \prod_{i=1}^{k} |V(\xi_i)| \cdot t(G[\xi_i]).$$

The number of distinct partitions reachable from ξ by the group action is:

$$|\mathrm{Pic}_k(G)| \,/\, |\mathrm{Stab}(\xi)|.$$

If the action of $\mathrm{Pic}_k(G)$ on rooted k-forests is transitive, this quotient equals the total number of connected k-partitions (with the given root set).

Warning: *The stabiliser is only defined if $\mathrm{Pic}_k(G)$ has a proven group action on rooted k-forests. Currently this action is conjectured (based on the counting match $|\mathrm{Pic}_k(G)|$ = number of rooted k-forests) but not proven. Without the action, $\mathrm{Stab}(\xi)$ is undefined and the formula $|\mathrm{Stab}(\xi)| = \prod |V(\xi_i)| \cdot t(G[\xi_i])$ is a heuristic based on fibre counting, not a theorem. Proving this action (e.g., via a multi-sink Bernardi bijection or multi-sink rotor-routing torsor) is the central open problem.*

Embedding into $\mathrm{Pic}^0(G)$**.** *A rooted k-forest with roots $\{q_1, \ldots, q_k\}$ (where $q_1 = q$ is the original sink of $\mathrm{Pic}^0(G)$) corresponds to a divisor in $\mathrm{Pic}^0(G)$ where the coordinates at $q_2, \ldots, q_k$ are zero. Define the embedding $\iota : \mathrm{Pic}_k(G) \hookrightarrow \mathrm{Pic}^0(G)$ by:*

$$\iota(D)_v = \begin{cases} D_v & \text{if } v \notin \{q_2, \ldots, q_k\}, \\ 0 & \text{if } v \in \{q_2, \ldots, q_k\}. \end{cases}$$

The image of ι is the subgroup of $\mathrm{Pic}^0(G)$ where the coordinates at $q_2, \ldots, q_k$ vanish. The surjection $\pi : \mathrm{Pic}^0(G) \to \mathrm{Pic}_k(G)$ (forgetting these coordinates) is the left inverse: $\pi \circ \iota = \mathrm{id}$.

Warning: *The map ι is NOT a well-defined group homomorphism. Equivalence in $\mathrm{Pic}_k(G)$ allows chips to vanish at the $k-1$ extra sinks $q_2, \ldots, q_k$. In $\mathrm{Pic}^0(G)$, those vertices are present and chips do not vanish. So $D \sim D'$ in $\mathrm{Pic}_k(G)$ does NOT imply $\iota(D) \sim \iota(D')$ in $\mathrm{Pic}^0(G)$. The map ι acts on representatives, not on equivalence classes. The surjection $\pi : \mathrm{Pic}^0(G) \to \mathrm{Pic}_k(G)$ IS well-defined (projecting respects equivalence because $\mathrm{im}(\Delta^{(q_1)})$ projects onto a superset of $\mathrm{im}(\Delta^{(\{q_1,\ldots,q_k\})})$), but the reverse embedding does not exist as a group homomorphism.*

The nested structure. *This gives a clean hierarchy:*

$$\{k\text{-partitions}\} \subset \{k\text{-forests}\} \subset \mathrm{Pic}_k(G) \hookrightarrow \mathrm{Pic}^0(G) \longleftrightarrow \{\text{spanning trees}\}.$$

Warning: *This diagram mixes rigorous and conjectured relationships. The $\longleftrightarrow$ (Baker–Wang torsor) is proven. The $\hookrightarrow$ is NOT a valid group homomorphism (see warning above). The second $\subset$ (k-forests $\subset \mathrm{Pic}_k(G)$) assumes a bijection between $\mathrm{Pic}_k(G)$ and rooted k-forests, which is a counting match ($|\mathrm{Pic}_k(G)|$ = number of rooted k-forests via Matrix-Forest Theorem) but NOT a proven group action or torsor. The first $\subset$ (k-partitions $\subset$ k-forests) is a set-level quotient, not an algebraic inclusion.*

At each level:

- $\mathrm{Pic}^0(G)$ *has $t(G)$ elements, indexing spanning trees.*
- $\mathrm{Pic}_k(G)$ *has $\det(\Delta^{(\{q_1,\ldots,q_k\})})$ elements, indexing rooted k-forests.*
- *k-forests modulo within-district equivalence give k-partitions.*

Each inclusion has computable index. The group structure of $\mathrm{Pic}^0(G)$ governs the entire hierarchy, and the partition structure is identified by the stabiliser at each level.

Remark. *Note that $\mathrm{Stab}(\xi)$ depends on ξ, so different partitions have different stabilisers. However, since $\mathrm{Pic}_k(G)$ is abelian, every subgroup is normal, and the quotient $\mathrm{Pic}_k(G)/\mathrm{Stab}(\xi)$ is a well-defined group for each ξ. The key open question is whether this quotient is "the same" (isomorphic) for all ξ, which would give a universal "partition Picard group" independent of the specific partition.*

22 Literature Review: Key Papers

22.1 Torsor Structures on Spanning Trees

Baker and Wang [4] prove that for planar graphs, the Bernardi process defines a *simply transitive* action of $\mathrm{Pic}^0(G)$ on spanning trees (Theorem 4.1). This action is canonical for planar graphs (independent of root vertex, Theorem 5.1) and coincides with the rotor-routing torsor (Theorem 7.1). This is the foundational result for the algebraic approach to spanning trees.

Backman, Baker, and Yuen [3] extend this to *regular matroids*: the Jacobian of a regular matroid admits a canonical simply transitive action on circuit-cocircuit reversal classes. The bijections β are constructed via acyclic signatures.

Ding, McDonough, Tóthmérész, and Yuen [11] prove that the sandpile torsor on regular matroids is *consistent* under deletion and contraction. This is the key property for our setting: contracting within-district edges gives the quotient graph Q_ξ, and consistency means the torsor on G projects to a torsor on Q_ξ.

22.2 Bijections for Spanning Subgraphs

Ding [10] extends the BBY bijection from spanning trees to *all spanning subgraphs*. Theorem 1.2 gives specialisations:

1. Spanning subgraphs $\leftrightarrow$ all orientations (cardinality $T_G(2,2)$).
2. **Spanning forests** $\leftrightarrow$ σ-compatible orientations (cardinality $T_G(2,1)$).
3. Connected spanning subgraphs $\leftrightarrow$ σ^*-compatible orientations (cardinality $T_G(1,2)$).
4. Spanning trees $\leftrightarrow$ (σ,σ^*)-compatible orientations (cardinality $T_G(1,1)$).

Row 2 is critical for us: spanning forests (= partitions into connected subgraphs) biject with σ-compatible orientations. However, the group action (torsor) is only established at the spanning tree level (row 4), not at the forest level.

22.3 Partitions and Divisors

Xu [22] establishes a direct map from ordered k-partitions to divisors. For an ordered partition $S = (V_1, \ldots, V_k)$, the *minimal effective divisor* D^S has value at each vertex $v \in V_i$ equal to the number of neighbours of v in districts V_j with $j > i$ (Proposition 2). Key consequences:

- Chip-firing $\chi(V_1)$ rotates the ordered partition: $\chi(V_1) \cdot D^S = D^{S^{(1)}}$ (Corollary 4).
- Merging adjacent districts $V_i \cup V_{i+1}$ corresponds to subtracting the boundary divisor: $D^S - D^{S'} = \sum_{v \in V_i} |\{vw \in E : w \in V_{i+1}\}| \cdot v$ (Corollary 3).
- The minimal free resolution of the toppling ideal R/I is (conjecturally) indexed by connected partitions (Conjecture 13, due to Wilmes).

22.4 The Matrix-Forest Theorem and $W(G) = I + L(G)$

Chebotarev and Shamis [7] prove the Matrix-Forest Theorems for the matrix $W(G) = I + L(G)$:

- $\det(W(G))$ = total number of rooted spanning forests (Theorem 5).
- The (i, j)-cofactor W^{ij} = number of rooted forests where i and j belong to the same tree rooted at i (Theorem 6).
- W_{ij}^{-1} = relative forest-accessibility: the fraction of forests where i and j are in the same tree rooted at j (Theorem 7).

The inverse W^{-1} is a *partition proximity matrix*: vertices with large W_{ij}^{-1} naturally belong in the same district across random forests. However, $W(G)$ counts forests of *all* component numbers simultaneously and does not directly encode k-partitions.

The referenced papers provide additional structure:

- **Bapat and Constantine** [5] count spanning forests with colour restrictions — potentially useful for encoding within-district vs. cross-boundary edge constraints algebraically.
- **Kelmans and Chelnokov** [14] show that the coefficients of the Laplacian characteristic polynomial equal the numbers of spanning forests with fixed numbers of components. This connects the *spectrum* of Δ_G to forest/partition counts at every level k simultaneously.
- **Merris** [16] proves an edge version of the Matrix-Tree Theorem using the edge Laplacian $B^T B$, potentially enabling an edge-based analogue of $\mathrm{Pic}_k(G)$.
- **Moon** [19] studies the adjoint of $W(G)$, which by Theorem 7 of [7] gives the matrix of relative forest-accessibilities.

22.5 Redistricting MCMC

DeFord, Herschlag, and Mattingly [8] introduce the Cycle Walk for sampling spanning forests for redistricting — the closest existing work to our approach.

Autry, Carter, Herschlag, Hunter, and Mattingly [2] develop multi-scale merge-split MCMC, addressing the same scaling problem as our boundary cycle step but via hierarchical proposals rather than algebraic structure.

McWhorter and DeFord [17] introduce the Marked Edge Walk, whose within-fibre waste motivates our entire programme.

23 Good News: What Is Now Proven (July 2026 Update)

This section records the outcome of a verification pass (July 2026) over the conjectures of Sections 14–21, combining literature that settles them with exhaustive machine checks on small graphs (the triangle, C_4, K_4, and the 2×3 and 3×3 grids). Headline: *the central conjecture of Section 21 is true*, and it follows from known results by a two-line reduction;

the multi-sink burning picture of Section 15 is exactly right; the fibre decomposition and Xu's divisor calculus are verified; and Wilmes' conjecture cited in Section 22.3 has been a theorem since 2015. (One caution so that this section is self-contained: the surjection $\pi : \mathrm{Pic}^0(G) \to \mathrm{Pic}_k(G)$ claimed in Section 16 is *false* — for the triangle with $k = 2$ we have $|\mathrm{Pic}_2| = 2 \nmid 3 = t(G)$, and 162 such divisibility failures were found across the test graphs — but none of the results below depend on it.)

23.1 The torsor on rooted k-forests exists

Theorem 1 (Multi-sink torsor via the quotient graph)**.** *Let G be connected, let $S = \{q_1, \dots, q_k\} \subseteq V$, and let G_S denote the (multi)graph obtained from G by identifying all vertices of S into a single vertex s^* and deleting the loops arising from edges inside S. Then:*

(i) the matrix $\Delta^{(\{q_1,\dots,q_k\})}$ of Section 14 is *the reduced Laplacian of G_S at the sink s^*; consequently $\mathrm{Pic}_k(G) \cong \mathrm{Pic}^0(G_S)$, with identical Smith normal forms;*

(ii) identification of the roots is a bijection between spanning forests of G rooted at S (each component containing exactly one q_i) and spanning trees of G_S;

(iii) rotor-routing on G_S with sink s^*, in the sense of Holroyd, Levine, Mészáros, Peres, Propp, and Wilson [13], defines a free and transitive *action of $\mathrm{Pic}_k(G)$ on the set of rooted k-forests of G. No planarity assumption is needed.*

Part (i) is a direct computation: non-root vertices keep their degrees in G_S (edges into S become edges into s^*), so deleting the k rows and columns of S from Δ_G produces exactly the reduced Laplacian of G_S. Part (ii) is the classical rooted-forest/contraction bijection. Part (iii) is the sandpile torsor theorem of [13] applied to the multigraph G_S.

This **resolves affirmatively the central open problem flagged in Section 21**: $\mathrm{Pic}_k(G)$ does act simply transitively on rooted k-forests. It also realises the multi-sink constructions of Section 15 verbatim: Dhar's burning algorithm on G_S started from s^* is precisely the "burn from k sinks simultaneously" Voronoi process, and the k-sink sandpile is the ordinary sandpile of G_S. Risk 2 of Section 20 is settled in its rotor-routing form. The Bernardi (planar-canonical) form genuinely fails in general, because G_S is usually non-planar even when G is planar: off planarity the torsor depends on the base data (Shokrieh and Wright [20]), and by the rigidity theorem of Ganguly and McDonough [12] — rotor-routing induces the *only* consistent sandpile torsor structure on plane graphs — there is no alternative consistent torsor to design around this.

Machine verification. The rotor action was implemented on the merged multigraphs and checked exhaustively: on the 2×3 grid with S a pair of opposite corners (21 rooted forests) and with a triple root set (13), on the 3×3 grid with opposite corners (288), and on K_4 with an adjacent root pair (so that G_S has parallel edges; 8). In every case each generator acts as a permutation of the trees of G_S, the chip-firing relations hold (so the action factors through $\mathrm{Pic}_k(G)$), the action is abelian, and a single orbit has full size — freeness plus transitivity, matching $\det \Delta^{(S)}$ exactly. The Matrix-Forest identities $\det \Delta^{(S)} = \#\{\text{rooted } k\text{-forests}\} = t(G_S)$ and the Smith normal form isomorphism of (i) were confirmed for *every* root set of every test graph.

23.2 Algorithmic corollaries

Corollary 1. *Fix the root set S. Then:*

(a) ***Exact sampling.*** *A uniformly random rooted k-forest of G is obtained by running Wilson's algorithm [21] on G_S with root s^*, in expected polynomial time. No Markov chain on forests, and no mixing-time argument, is needed for the unconstrained problem.*

(b) ***Navigation.*** *The Smith normal form of $\Delta^{(S)}$ gives coordinates on $\mathrm{Pic}_k(G)$; with Theorem 1(iii) the group element carrying any rooted forest to any other is well defined and computable in polynomial time. This makes the navigation algorithm of Section 19 rigorous.*

(c) ***Exact fibre sampling.*** *For a fixed connected k-partition ξ compatible with S, the fibre of rooted forests inducing ξ has exactly $\prod_i t(G[\xi_i])$ elements, and it can be sampled exactly: Wilson's algorithm inside each district, and a uniform element of $\mathrm{Pic}^0(Q_\xi)$ of the weighted quotient multigraph converted by Dhar's burning on Q_ξ into a uniform boundary configuration.*

23.3 The fibre decomposition is verified

The counting identity $\tau(\xi) = \left(\prod_i t(G[\xi_i])\right) \cdot t_w(Q_\xi)$ for the number of spanning trees compatible with ξ was verified by brute-force enumeration on five partitions of the 2×3 and 3×3 grids (values $3, 4, 8, 45, 32$), in addition to the 6×6 computation in the companion note. Two remarks. First, the identity is Cauchy–Binet counting and needs no functoriality; this matters because *no* group homomorphism exists between $\mathrm{Pic}^0(G)$ and $\mathrm{Pic}^0(Q_\xi)$ in general (triangle with $\xi = \{1,2\} \mid \{3\}$: $t_w(Q_\xi) = 2 \nmid 3 = t(G)$). Second, the $\mathrm{Pic}^0(Q_\xi)$ factor is exactly the boundary-configuration count, which is what makes Corollary 1(c) an *exact* within-fibre resampler — to our knowledge the first such for redistricting chains.

23.4 Irreducibility: the unconstrained statement survives testing

The boundary-restricted single-swap chain (cycle steps with e^+ restricted to boundary non-tree edges, marked edges free) was enumerated exhaustively on the 3×3 grid with $k = 2$: all 1,536 states (T, M) form a *single* communicating class covering all 53 connected 2-partitions. The irreducibility statement is therefore plausible as a theorem on nice graph families, and two published tools point at a combinatorial proof: Ganguly–McDonough's Corollary 5.16 [12] (spanning trees of any 2-connected graph are connected under single exchanges that remove only *leaf* edges) and the column-canonicalisation argument of Akitaya et al. [1] (Theorem B.2: irreducibility with *exact* balance for $k = 2$ on simple grid graphs, with diameter $O(n)$). Under exact balance the same enumeration ($k = 3$, districts of size 3) fractures at the state level (two isolated states), consistent with the non-irreducibility example on grids with holes in [1]; unconstrained and balance-restricted irreducibility must be treated as separate theorems.

23.5 Xu's divisor calculus is verified; Wilmes' conjecture is a theorem

Xu's Proposition 2 (existence, legality, *and* minimality of D^S), Corollary 3 (merge formula), and Corollary 4 (rotation) were verified on all 1,900 ordered connected partitions of the three

test graphs ($k = 2, 3$), with no exceptions. Note also $\deg D^S = |\partial\xi|$: the degree of Xu's divisor is the cut size of the partition.

Conjecture 13 quoted in Section 22.3 is no longer a conjecture: **Manjunath, Schreyer, and Wilmes** [15] and, independently, **Mohammadi and Shokrieh** [18] proved that the minimal free resolution of the toppling ideal is supported on connected partitions exactly as predicted. Connected k-partitions therefore carry a *proven* global algebraic structure — they index the $(k-1)$-st syzygies of the toppling ideal, with Xu's D^S as the multidegrees and merge/refinement as the differentials. One structural limitation should be recorded alongside the good news: chip-firing preserves divisor classes, and the classes $[D^S]$ fragment partition space (on the 3×3 grid with $k = 2$: 53 partitions spread over 49 classes), so Xu's dynamics rotates labels within a partition (Corollary 4) rather than exploring partitions — the partition-changing moves are exactly the *non-principal* merge divisors of Corollary 3. Read positively: every merge–split (ReCom-style) move has an exact algebraic ledger with effective certificates.

23.6 Consequences for the programme

The algebraic layer now stands on proven ground: exact uniform sampling of rooted k-forests (no mixing proof needed), polynomial-time navigation, an exact within-fibre resampler, and a torsor that quotients away the tree-level redundancy. What the algebra provably does not see is the balance constraint (partition fibres are not cosets of subgroups of $\mathrm{Pic}_k(G)$; verified on the 2×3 grid), which is where the geometric results enter: Cannon, Pegden, and Tucker-Foltz [6] prove that balanced k-forests are a $1/\mathrm{poly}$ fraction of all k-forests on grids and obtain the first provably polynomial-time exact sampler of balanced tree-weighted partitions there. The designated target of the programme is to combine the exact fibre machinery above with such geometric bounds into a mixing proof for a balance-restricted chain on grids — Configuration B (Remark 8.2) of the companion note — which no chain currently possesses.

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